

Recruitment Funnel Analytics — People Analytics Project 2023–2025.

A full people analytics project | 1,000 applicants | Jan 2023 – Mar 2025

10 departments · 6 source channels · 30 hiring managers

This notebook analyses a complete recruitment dataset from application to joining. Each section surfaces a different layer of the hiring story — where candidates drop, which channels perform, where time is lost, and which hiring managers breach SLAs. Built as part of a people analytics portfolio project using Google Sheets, Airtable, Looker Studio, and Observable.

Data Sources

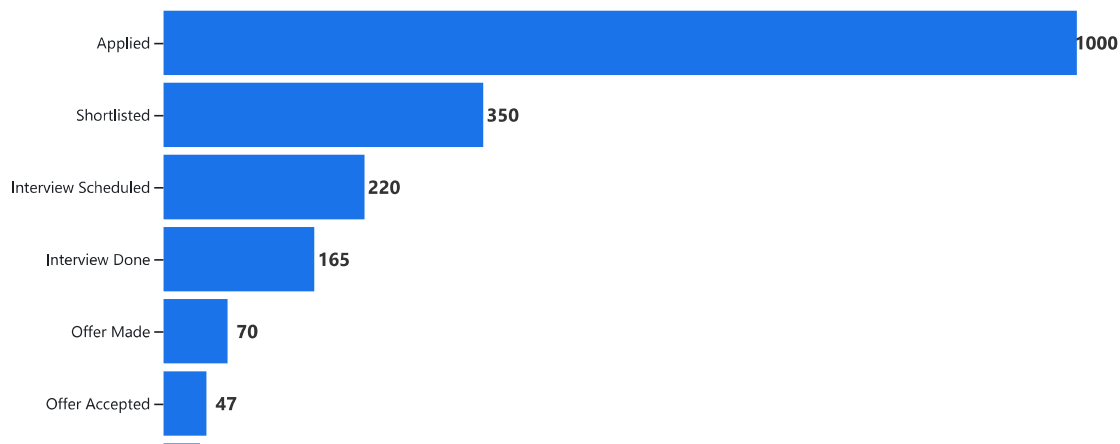
Five datasets power this notebook — one raw applicant file with 1,000 rows and 29 columns, and four pre-aggregated summary tables for funnel, channel, time-to-hire, and rejection analysis. All data originates from Google Sheets and was exported as CSV for Observable.

1. Recruitment Funnel — Where Candidates Drop

Every stage of the funnel loses candidates. The question is whether the loss is by design or by failure. A healthy funnel has controlled drop-off at screening and minimal drop-off after the offer stage. This chart shows cumulative reach — how many candidates made it to at least each stage.

Watch for: The steepest drop happens between Applied and Shortlisted. Only 35% of applicants clear the first screen. After that, interview-to-offer is the next major loss point at 44.9% conversion — candidates are making it through interviews but not getting offers, which points to either indecision at the hiring manager level or a misaligned expectation on both sides.

Recruitment Funnel — Cumulative Candidates at Each Stage





2. Source Channel ROI — Volume vs Quality vs Speed

Not all sourcing channels are equal. LinkedIn dominates volume but that means nothing if the conversion rate is low. The real question is: for every 100 applications a channel sends, how many become hires, and how fast?

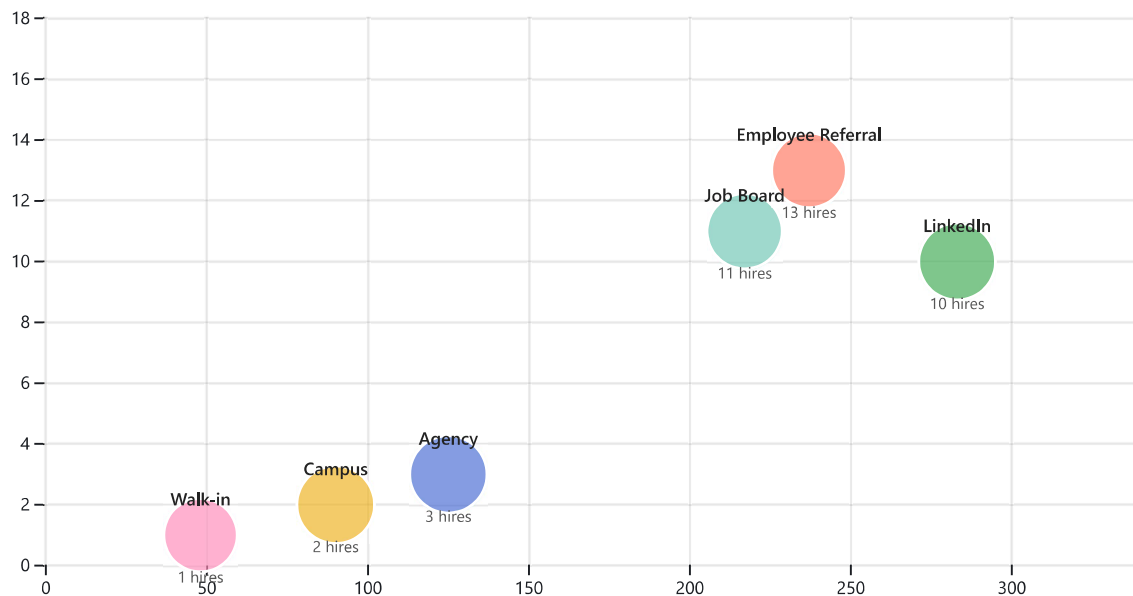
This chart shows all three dimensions simultaneously. X axis is volume, Y axis is hires, bubble size is average time to hire. The ideal channel sits high on Y, moderate on X, with a small bubble.

Watch for: Employee Referral sits closest to the ideal position. LinkedIn sits far right with a large bubble — maximum volume, below-average conversion, slowest average TTH at 58 days. Walk-in is the fastest channel at 50 days but produces minimal volume. Campus Recruitment has the worst conversion at 2.2% despite being a deliberate investment channel.

Source Channel ROI — Volume vs Quality vs Speed

Bubble size = Avg Time to Hire. Higher up = more hires.

↑ Candidates Joined



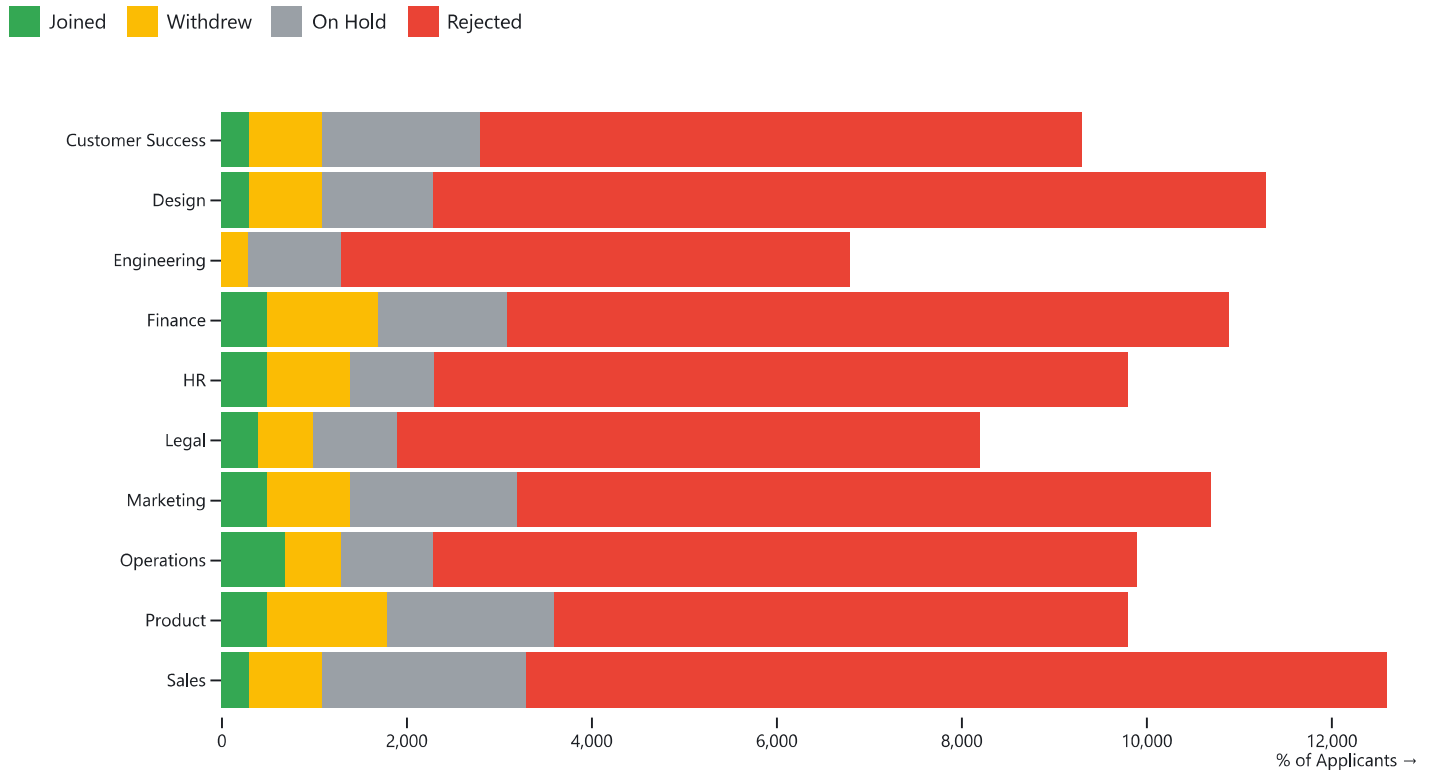
Total Applications →

3. Candidate Outcomes by Department

Every department runs its own hiring process through the same central recruitment function. But outcomes vary dramatically. Some departments are converting applicants consistently. Others are accumulating On Hold candidates which signals internal indecision, budget freezes, or hiring managers who are not prioritising the process.

Watch for: Engineering is the standout problem — 69 applicants, zero hires. The entire bar is Rejected and On Hold. This is not a sourcing failure. It is either a budget freeze mid-pipeline, a compensation mismatch, or interview standards set too high relative to the candidate pool available. Every other department has at least some green. Engineering has none.

Candidate Outcomes by Department



4. Hiring Manager SLA Compliance — Speed by Stage

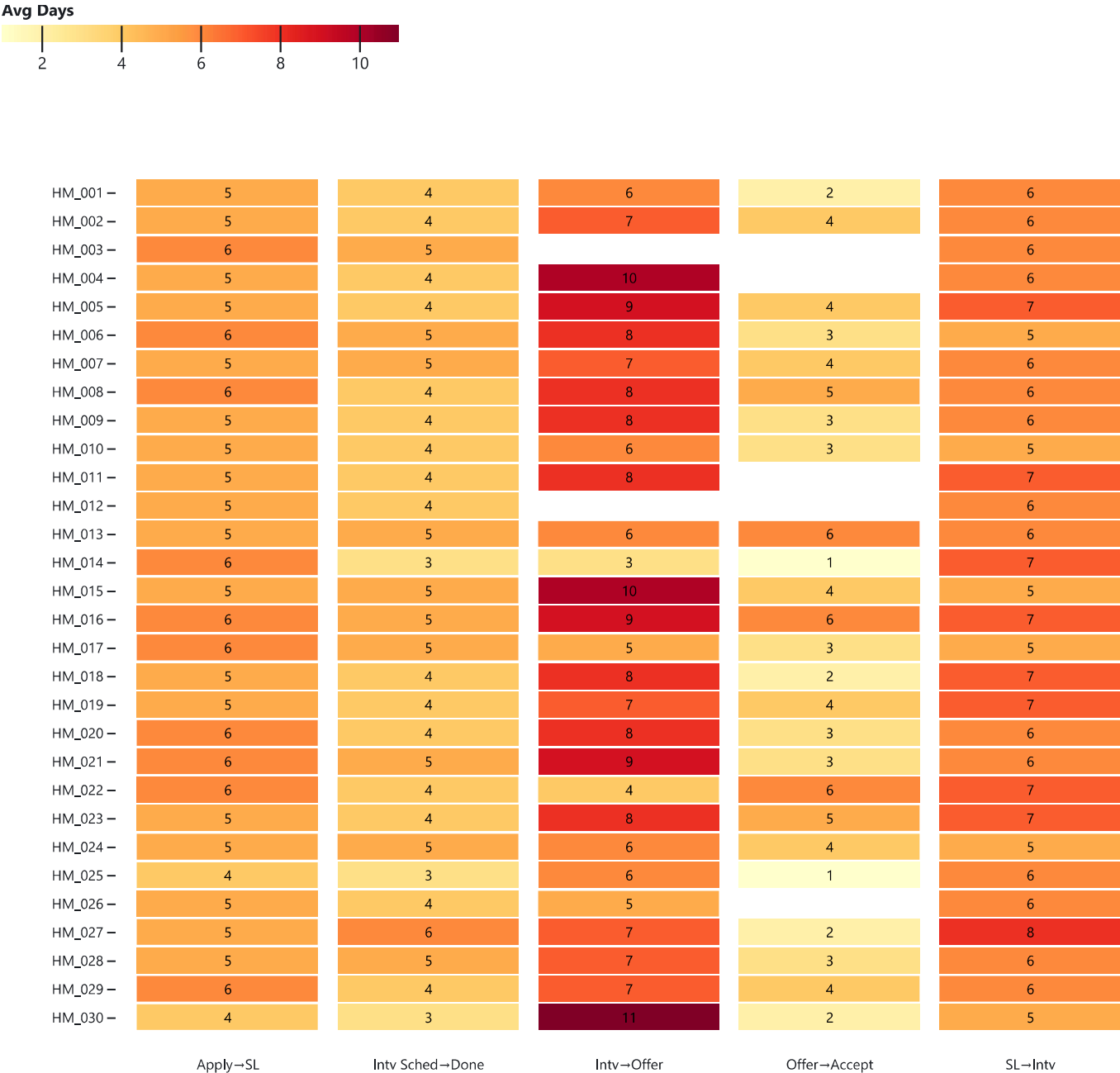
Time to hire is not just a recruitment problem. Hiring managers control the most time-sensitive stages — scheduling interviews, giving feedback, and approving offers. When a hiring manager sits on a shortlist for 12 days instead of 5, the best candidates accept competing offers.

This heatmap shows average business days each hiring manager takes at each stage. White cells are fast. Deep red cells are SLA breaches. The pattern reveals whether slowness is systemic across the whole organisation or concentrated in specific managers.

Watch for: Any cell showing more than 7 days for Apply to Shortlist or more than 5 days for Interview to Offer is a breach of standard SLA. Managers with multiple red cells across stages are your biggest time-to-hire risk and the first conversation a head of TA should have before the next quarterly review.

Hiring Manager Speed by Stage (Avg Business Days)

Deeper red = slower. Use this to identify SLA breaches.

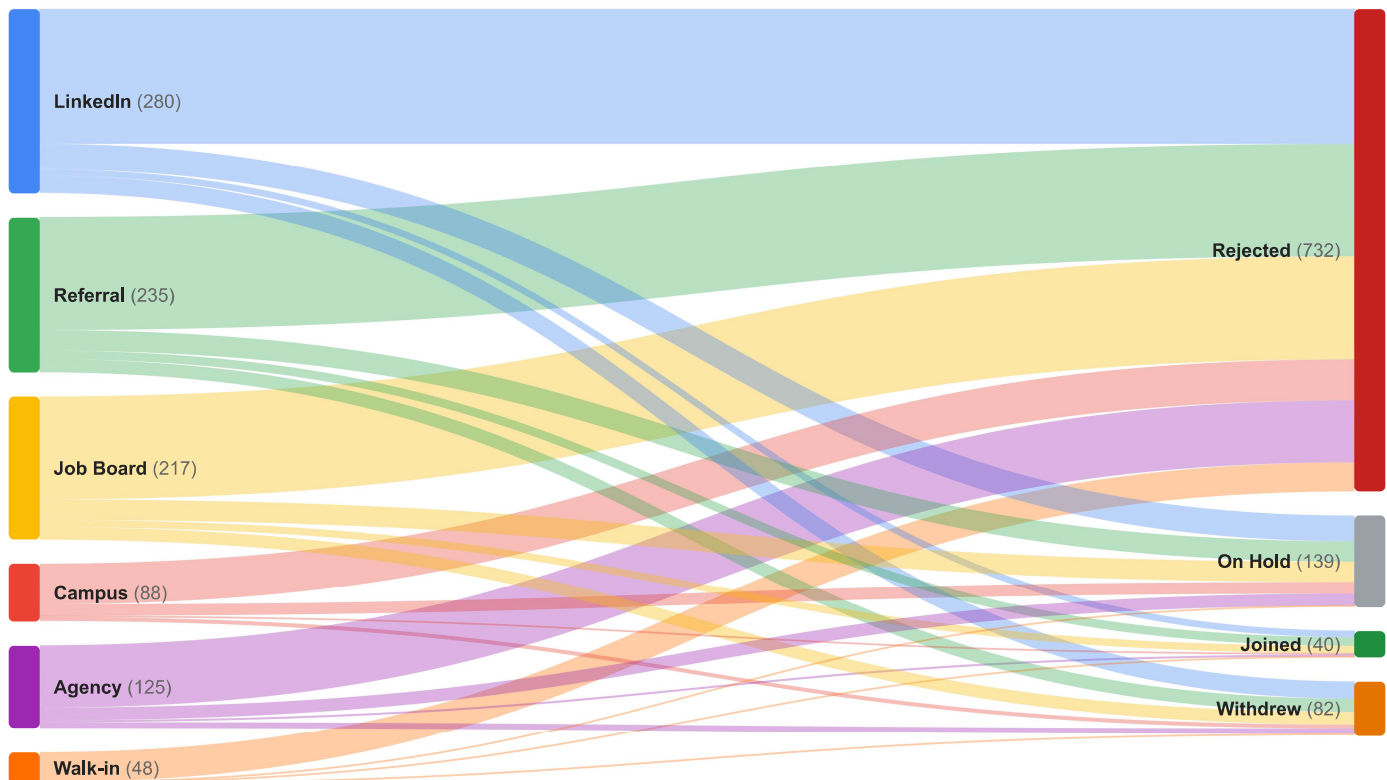


5. Candidate Flow — From Source to Outcome

The Sankey diagram shows the full candidate journey as flowing streams. Width represents volume — a wider stream means more candidates moving along that path. Colour represents the originating source channel. Reading left to right shows exactly where each channel's candidates end up.

Watch for: LinkedIn's stream starts as the widest on the left but the proportion flowing into Joined on the right is smaller than its input width suggests. Employee Referral's stream is narrower at entry but a larger share reaches Joined proportionally. The Rejected outcome node on the right is by far the widest — 732 candidates, dominated by LinkedIn and Job Board inputs. On Hold at 139 is disproportionately large for a healthy pipeline.

Candidate Flow: Source Channel → Final Outcome



6. Application Volume by Quarter

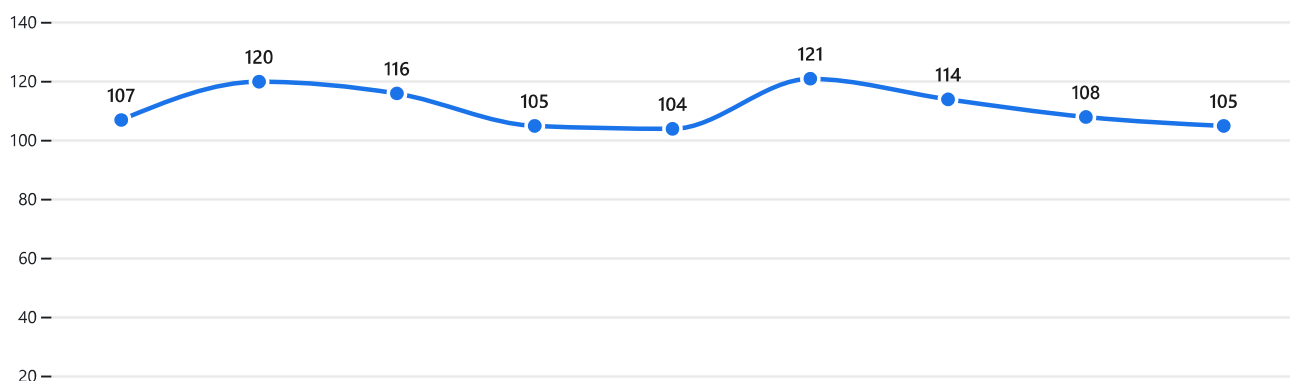
Hiring activity is rarely constant. Volumes spike in certain quarters due to business growth, attrition waves, or annual headcount approvals. Understanding seasonality helps recruitment teams plan capacity — when to ramp up sourcing, when to expect pressure on interview scheduling, and when hiring will naturally slow.

Watch for: Any quarter with a sharp spike in applications without a corresponding spike in hires indicates a sourcing quality problem — volume came in but the pipeline did not convert. A flat or declining trend in later quarters may reflect budget freezes, which aligns with the 77 Budget Frozen rejection reasons in the dataset.

Application Volume by Quarter

Hiring activity and seasonality across 2023–2025

↑ Applications



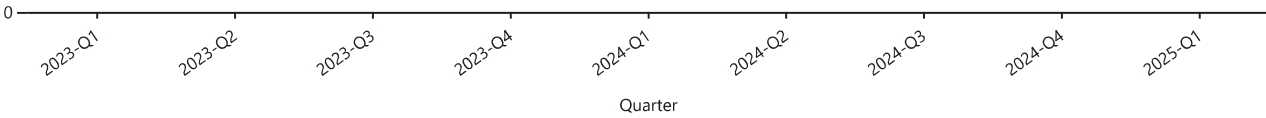


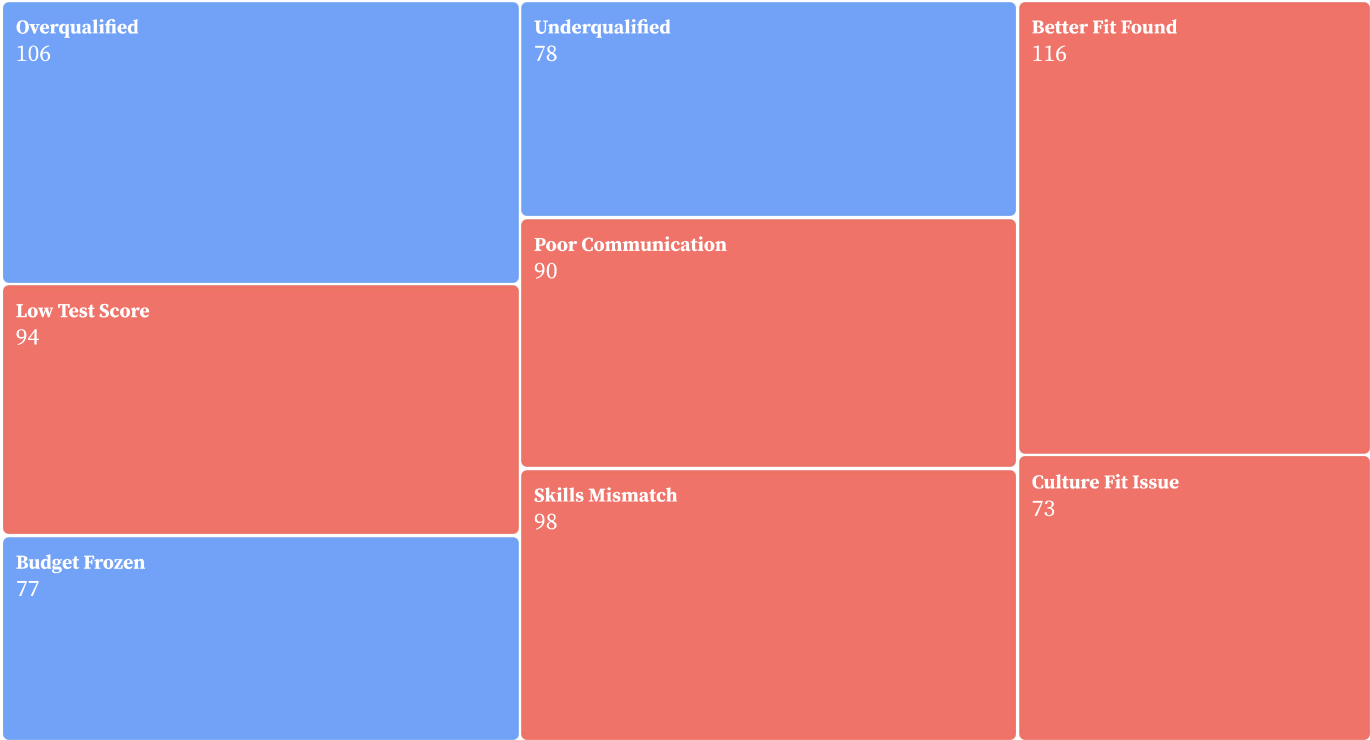
Chart 7: Rejection Treemap 7. Why Candidates Are Rejected — Treemap Analysis

Rejection reasons are the most underanalysed data point in recruitment. They reveal whether your screening is working, whether your job descriptions are attracting the right people, and whether internal decisions are costing you good candidates.

This treemap sizes each rejection reason by frequency. Larger rectangle means more rejections under that reason. Red rectangles are company-controlled — things the organisation can fix. Blue rectangles are external — harder to control directly.

Watch for: Better Fit Found at 116 is the top reason and is broadly healthy — it means selection is happening. But Skills Mismatch at 98 and Low Test Score at 94 together suggest the sourcing channels are not filtering well enough before application, or the job descriptions are unclear about requirements. Budget Frozen at 77 is a planning failure not a recruitment failure — those 77 candidates entered a pipeline that was never going to result in a hire.

Rejection Reasons — Treemap (larger = more frequent)



Summary of Key Findings

After analysing 1,000 applicants across 10 departments and 6 source channels, these are

the five findings that matter most from a business perspective.

- 1. Overall hire rate is 4%** — 40 people joined from 1,000 who applied. Industry benchmark for a healthy funnel is 3–8% so the rate itself is acceptable. The composition of losses is where the problems lie.
- 2. Engineering hired nobody** — 69 applicants, zero joins. This is the single most important finding and needs immediate investigation before the next hiring cycle opens.
- 3. On Hold at 13.9% is three times the healthy benchmark** — 139 candidates are sitting in limbo. This is wasted recruiter effort and damaged employer brand. Every On Hold candidate is a potential future Withdraw.
- 4. Employee Referral is your best channel** — highest conversion at 5.5%, fastest average TTH at 52.5 days, proportionally strongest flow to Joined in the Sankey. The referral programme deserves more investment relative to LinkedIn spend.
- 5. Budget Frozen accounts for 77 rejections** — candidates entered a pipeline that had no chance of resulting in a hire. This is a requisition planning problem, not a recruitment problem, and it sits with business leadership not the TA team.

Built with Google Sheets · Gemini in Sheets · Airtable · Looker Studio · Observable Data: Mock dataset | 1,000 applicants | Jan 2023 – Mar 2025


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d3Sankey = require("d3-sankey@0.12");
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